B.Tech Minor Project Report

Semi-Supervised Dental Panoramic Caries Segmentation Using Multi-Level Uncertainty Aware Learning

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Abstract

Dental caries (tooth decay) is one of the most prevalent oral diseases globally, and its early detection from panoramic radiographs is critical for timely clinical intervention. This project presents a re-implementation and extension of the Multi-Level Uncertainty Aware (MLUA) semi-supervised segmentation framework proposed by Wang et al. [1], applied to the DC1000 dataset of dental panoramic X-ray images. The core challenge lies in distinguishing genuine carious lesions from imaging artifacts in the presence of limited labeled data and extremely small foreground regions. Our implementation adopts a ResNet34-based Feature Pyramid Network (FPN) decoder with attention gates and deep supervision, trained within a mean-teacher semi-supervised framework. We experimented with multiple loss functions (combined BCE+Dice, Tversky, and Focal-Tversky losses), Monte Carlo uncertainty sampling, and various hyperparameter configurations. Our best configuration achieved a test Dice score of **0.6619** and an IoU of **0.5492**, demonstrating the effectiveness of multi-level uncertainty aware consistency regularization for challenging small-target medical segmentation tasks.

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1.1 Motivation

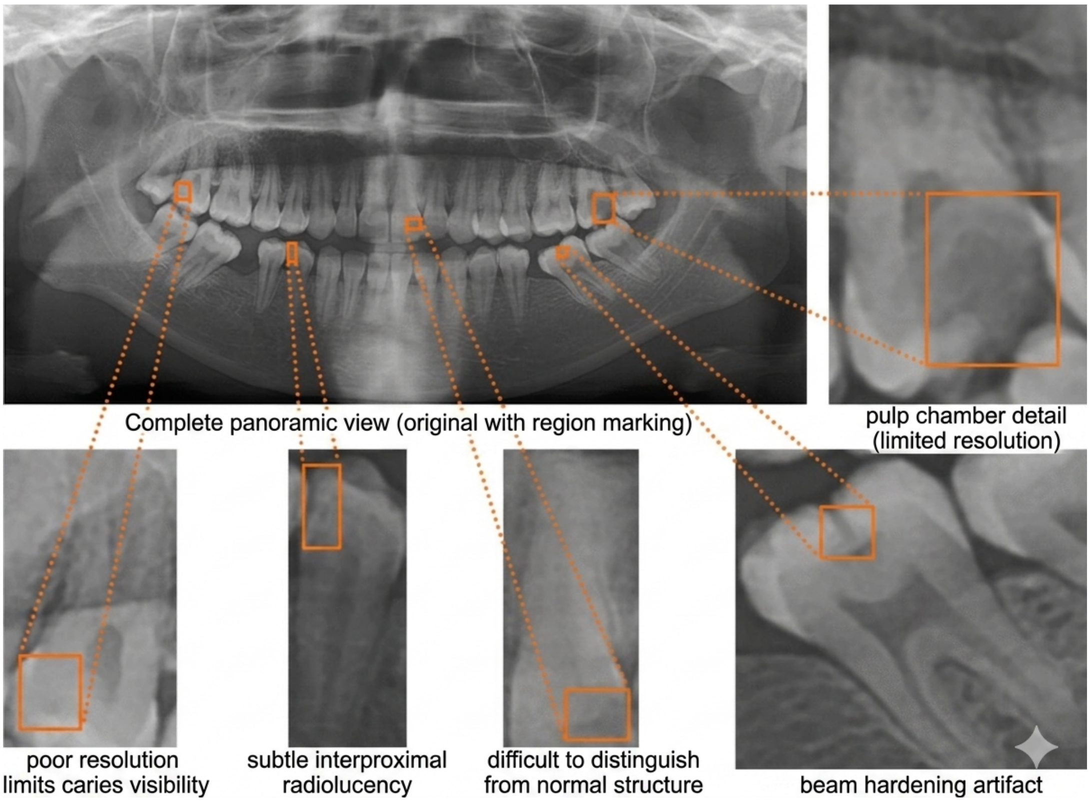

Figure 1: Challenges in dental panoramic radiograph interpretation for caries segmentation. The framework addresses several key difficulties: the global context in the Complete panoramic view, poor resolution limiting caries visibility, subtle interproximal radiolucency, regions difficult to distinguish from normal anatomy, beam hardening artifacts, and the need for high-resolution pulp chamber detail.

Dental caries is a chronic infectious disease that progressively destroys tooth structure. According to the World Health Organization, it stands alongside cardiovascular disease and cancer as one of the three most important diseases requiring prevention and treatment. Despite its high global prevalence, the treatment rate remains significantly low, particularly in developing nations, owing to limited access to experienced dental professionals.

Clinical diagnosis of dental caries primarily relies on oral panoramic radiographs, which provide a comprehensive overview of the entire dentition in a single image. However, accurate interpretation of these images is heavily dependent on the clinical experience of the examining dentist, introducing significant inter-observer variability and leading to missed or incorrect diagnoses.

Computer-aided diagnosis (CAD) systems powered by deep learning offer a promising avenue to standardize and assist clinical decision-making. Yet, developing effective segmentation models for caries in panoramic X-rays presents several unique challenges:

1. **Extreme class imbalance:** Caries lesions on average occupy only 1.5 of the total panoramic image area, making it statistically easy for a model to predict all-background and still appear accurate.

2. **Artifact interference:** Panoramic X-ray images suffer from numerous artifacts caused by patient movement, equipment quality, and operator technique. These low-density shadows mimic the appearance of carious lesions, creating genuine diagnostic ambiguity.

3. **Scale diversity:** Caries lesions span a vast range of sizes, from tiny shallow enamel lesions (under 100 pixels) to large deep caries (over 4000 pixels), requiring models to be sensitive across multiple scales simultaneously.

4. **Annotation scarcity:** Producing accurate pixel-level annotations for caries requires significant expert effort and time. In many cases, even expert dentists cannot definitively classify a region as a lesion or artifact, making certain annotations genuinely unreliable.

These challenges collectively motivate the use of a *semi-supervised* approach that can leverage both the limited labeled data (with reliable annotations) and the larger pool of unlabeled or ambiguously-labeled data (regions where annotation is contested) to improve segmentation performance.

1.2 Objective

The primary objectives of this project are:

1. **Reproduce and understand** the Multi-Level Uncertainty Aware (MLUA) semi-supervised framework for dental caries segmentation as proposed by Wang et al. [1].

2. **Implement** the full pipeline including data preparation, the FPN-based segmentation backbone with deep supervision, the mean-teacher SSL framework, and Monte Carlo uncertainty-based consistency training.

3. **Experiment** with architectural enhancements, including attention gates, weighted feature merging, and improved FPN decoder designs, to study their impact on segmentation quality.

4. **Evaluate** different loss function strategies (BCE+Dice, Tversky, Focal-Tversky, combined Tversky+Focal) and their interaction with the semi-supervised consistency loss.

5. **Analyze** the effect of hyperparameters such as Monte Carlo sampling count (T), training epochs, EMA decay (α), uncertainty threshold (β), and consistency weight (w) on model performance.

6. **Achieve** competitive segmentation performance on the DC1000 dataset benchmark, approaching or exceeding the results of the baseline fully-supervised models.

2 Literature Review

2.1 Dental Caries Detection Using Deep Learning

Early work on automated caries detection relied on traditional image processing techniques. Singh et al. [11] applied Radon and Discrete Cosine Transforms on panoramic images to capture low-frequency caries signatures, while Na'am et al. [12] utilized morphological gradients for identifying proximal caries edges. These methods, lacking high-level semantic understanding, showed poor generalization to diverse clinical conditions.

The advent of deep learning significantly improved the state of caries detection. AL-bahbah et al. [13] used VGG-16 CNNs for classifying dental diseases. Lee et al. [14] demonstrated high accuracy using the Inception-V3 architecture on periapical radiographs. Vinayahalingam et al. [15] trained MobileNetV2 for caries classification in third molars. Most recently, Zhu et al. [10] introduced *CariesNet*, a U-shaped segmentation network with partial encoder modules and full-scale axial attention to delineate multi-stage caries from panoramic images. However, these methods focus on fully supervised settings and do not address the challenge of uncertain or ambiguous lesion regions.

2.2 Semi-Supervised Medical Image Segmentation

Semi-supervised learning (SSL) has attracted considerable attention in medical imaging, where annotation cost is high. The field broadly divides into pseudo-label based methods and consistency regularization approaches.

Pseudo-label methods iteratively generate labels for unlabeled data using a model trained on labeled samples, then retrain with the augmented set. Fan et al. [16] applied random pseudo-label sampling for COVID-19 lung segmentation, though at a significant computational cost.

Consistency regularization methods enforce prediction invariance under perturbations. Temporal Ensembling [3] produces consistent predictions across different training iterations using an accumulated ensemble. Mean Teacher [2] replaced this with a continuously EMA-updated teacher network that guides a student via consistency loss on unlabeled data, becoming a foundation for many subsequent methods.

The **Uncertainty Aware Mean Teacher (UAMT)** [4] extended this framework by using Monte Carlo (MC) Dropout to estimate prediction uncertainty on the teacher's outputs, applying consistency loss only in high-confidence (low-uncertainty) regions. This approach proved particularly effective for 3D left atrium segmentation.

The **Uncertainty Rectified Pyramid Consistency (URPC)** [5] further extended the idea by generating multi-scale pyramid predictions from the student network and using inconsistency between scales to define uncertainty, without requiring a separate teacher network.

The **MLUA** framework [1], which this project implements and extends, advances beyond these methods by combining three distinct perturbation strategies—iterative (EMA), noise (Gaussian), and multi-scale (deep supervision)—to generate a richer and more stable uncertainty mask via Monte Carlo integration. This multi-level approach is specifically suited to the challenges of panoramic caries segmentation.

2.3 Feature Pyramid Networks and Attention Mechanisms

Feature Pyramid Networks (FPNs) [6] enable multi-scale feature extraction by combining high-resolution, semantically weak features from early layers with low-resolution, semantically strong features from deep layers through a top-down pathway with lateral connections. FPNs are particularly well-suited for detecting objects across a wide range of scales, making them applicable to caries segmentation.

Attention gates, popularized in medical image segmentation by Oktay et al. [7], enable the network to selectively focus on diagnostically relevant regions while suppressing irrelevant background activations. Incorporating attention into FPN skip connections can improve sensitivity to small-scale lesions.

Deep supervision [17] enforces intermediate predictions at multiple decoder levels, improving gradient flow and encouraging multi-scale feature learning. In the MLUA context, deep supervision additionally multiplies the number of prediction samples available for Monte Carlo uncertainty aggregation at minimal extra cost.

3 Technical Details: Design and Implementation

3.1 Dataset: DC1000

The DC1000 dataset [1] is the first public dataset for dental panoramic caries segmentation. It contains 1000 panoramic X-ray images with pixel-level annotations for carious lesions across three severity classes: shallow, middle, and deep caries. Key dataset statistics are summarized in Table 1.

Table 1: DC1000 Dataset Statistics

Property	Value
Total panoramic images	1000
Fully annotated images	593
Rough annotation images	407
Total caries lesions (labeled set)	4,520
Average caries per image	7.6
Average foreground proportion	1.5
Image dimensions	2943×1435 pixels
Training patch size	384×384 pixels
Annotated training slices (patches)	1,460

3.1.1 Data Split Strategy

To replicate the paper's experimental conditions, we divided the dataset by image ID:

- **test set:** First 100 image slices (fixed, image IDs 1–100).
- **Labeled training set:** Image slices with IDs below 1558 (approximately 50% labeled ratio).
- **Unlabeled training set:** Remaining image slices with IDs ≥ 1558.

This ID-based split ensures that labeled and unlabeled data originate from different patient cases, preventing data leakage and replicating the clinical reality of having a mix of confidently annotated and ambiguous cases.

3.2 Experimental Setup and Hyperparameter Configuration

All experiments were conducted using the PyTorch deep learning framework on an NVIDIA GPU with mixed precision training enabled for computational efficiency. The implementation follows the semi-supervised mean-teacher paradigm with uncertainty-aware consistency regularization inspired by the original MLUA framework.

To ensure reproducibility and fair comparison across experiments, all architectural settings, optimization strategies, and data preprocessing operations were kept fixed unless explicitly varied for ablation analysis.

The primary variables and hyperparameters used throughout the experiments are summarized below.

Table 2: Experimental Variables and Hyperparameter Definitions

Symbol / Parameter	Description
E	Total number of training epochs
T	Number of Monte Carlo uncertainty sampling passes
α	Exponential Moving Average (EMA) decay factor for teacher model update
β	Uncertainty threshold coefficient controlling reliable pseudo-label selection
w	Maximum consistency regularization weight
λ	Dynamic consistency loss weight
t	Final probability threshold used during panorama inference
γ	Focal loss focusing parameter
α_{tv}	Tversky loss false positive weighting factor
β_{tv}	Tversky loss false negative weighting factor
η	Learning rate used by the AdamW optimizer
B	Total batch size
B_l	Number of labeled samples per batch
B_u	Number of unlabeled samples per batch
$H \times W$	Input image resolution
θ_f	Student model parameters
θ_g	Teacher model parameters
$m_{certain}$	Binary certainty mask generated from uncertainty estimation
$m_{uncertain}$	Pixel-wise uncertainty map

3.2.1 Data Augmentation

To improve generalization and avoid overfitting on the limited labeled set, we applied the following augmentations during training:

- **Random horizontal flip** (p = 0.5) applied identically to image and mask using a fixed random seed to preserve spatial correspondence.
- **Resize** to 384×384 pixels for uniform batch processing.
- **Normalization** with mean 0.5 and standard deviation 0.5, mapping pixel values to $[-1, 1]$.
- **Gamma augmentation** (during MC sampling): Random gamma correction with $\gamma \in [0.7, 1.5]$ simulates varying X-ray exposure conditions.
- **Additive Gaussian noise** (during MC sampling): $\mathcal{N}(0, 0.05^2)$, clipped to $[-0.2, 0.2]$, simulates equipment and patient-motion artifacts.

Table 3: Final Experimental Configuration

Hyperparameter	**Value**
Input resolution	384×384
Optimizer	AdamW
Learning rate (η)	3×10^{-4}
Weight decay	10^{-4}
Batch size (B)	8
Labeled samples per batch (B_l)	4
Unlabeled samples per batch (B_u)	4
Maximum epochs (E)	50
EMA decay (α)	0.99
Monte Carlo samples (T)	10
Consistency weight (w)	0.2
Uncertainty threshold coefficient (β)	0.75
Dropout rate	0.3
Focal loss γ	2.0
Tversky α_{tv}	0.3
Tversky β_{tv}	0.7
Weighted BCE positive weight	10.0
Inference threshold (t)	0.25
Precision mode	Mixed precision fp16

3.3 Overall Architecture

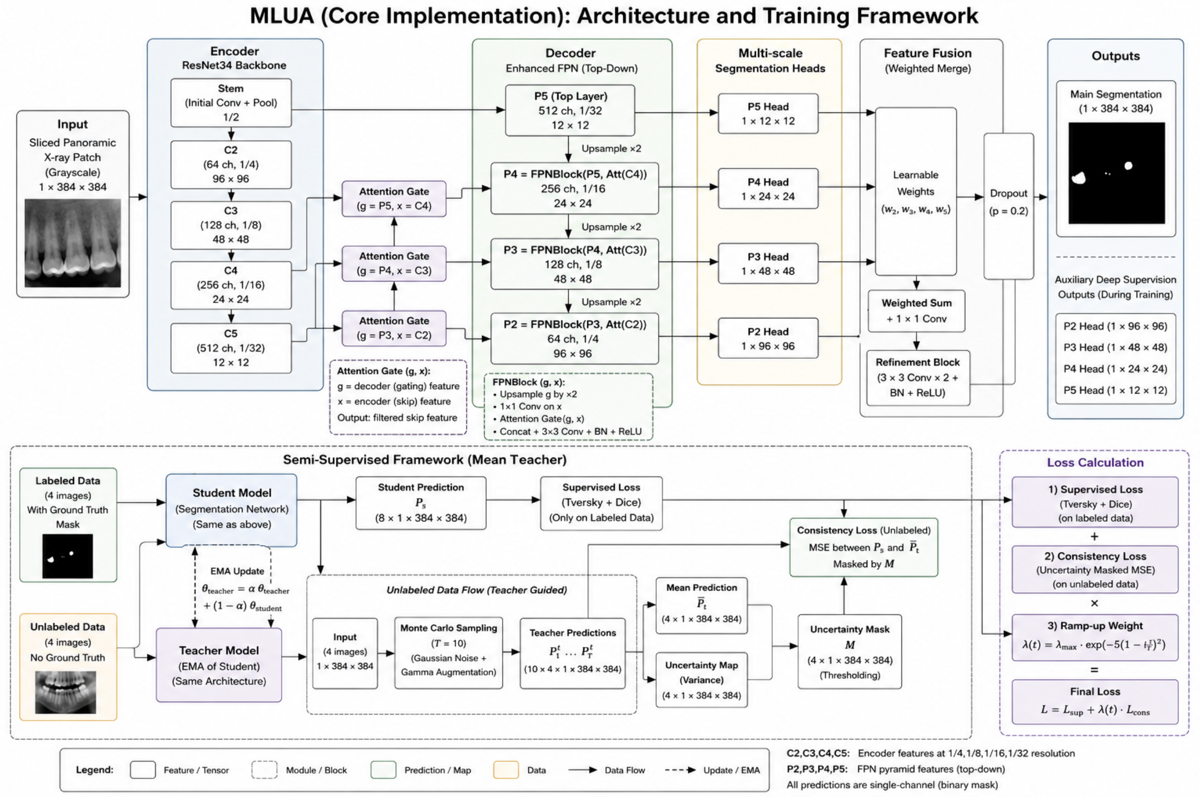

Figure 2: MLUA: Semi-Supervised Dental Caries Segmentation Network. The architecture integrates an attention-enhanced FPN decoder with a teacher–student semi-supervised framework and uncertainty-aware consistency learning.

The system follows a twin-network (mean-teacher) architecture, as illustrated in Figure 2. The student network is trained with both supervised loss on labeled data and consistency loss on unlabeled data. The teacher network maintains an Exponential Moving Average (EMA) of the student's parameters and generates uncertainty-filtered pseudo-targets for unlabeled data.

3.4 Backbone: ResNet34 Encoder

The encoder uses a **ResNet34** pre-trained on ImageNet, adapted for single-channel (grayscale) input by modifying the first convolutional layer to accept 1-channel images. The encoder outputs feature maps at five stages with progressively reduced spatial resolution and increased channel depth, denoted $\{C_1, C_2, C_3, C_4, C_5\}$.

3.5 FPN Decoder with Attention Gates

The decoder follows a Feature Pyramid Network design enhanced with two key improvements:

3.5.1 Attention Gates

Each lateral skip connection in the FPN incorporates an **Attention Gate (AG)**, adapted from the Attention U-Net design [7]. For a gating signal $\mathbf{g}$ from the deeper layer and skip feature $\mathbf{x}$ from the encoder:

$$\alpha = \sigma\big(\psi^\top(\mathrm{ReLU}(W_g\mathbf{g} + W_x\mathbf{x}))\big) \tag{1}$$

$$\hat{\mathbf{x}} = \alpha \odot \mathbf{x} \tag{2}$$

where σ denotes sigmoid activation, ψ, W_g, W_x are learned 1×1 convolutions, and $\odot$ is element-wise multiplication. The attention coefficient α soft-selects spatially relevant features, suppressing uninformative background activations.

3.5.2 Weighted Feature Merging

Instead of simple summation or concatenation of multi-scale segmentation features, we employ a **learnable weighted merge** module. Given segmentation features $\{f_1, f_2, f_3, f_4\}$ from four pyramid levels (after upsampling to a common resolution):

$$\mathbf{F}_{\text{merge}} = \sum_{i=1}^{4} w_i f_i, \quad w_i = \mathrm{Softmax}(\mathbf{v})_i \tag{3}$$

where $\mathbf{v} \in \mathbb{R}^4$ is a learnable weight vector. This allows the model to adaptively emphasize the most informative scale for different regions.

3.5.3 Feature Refinement and Dropout

After merging, two successive `Conv3×3-GroupNorm-ReLU` blocks refine the merged feature map, followed by spatial `Dropout2d` (rate 0.3) to prevent co-adaptation and improve uncertainty estimation quality.

3.5.4 Deep Supervision

Four auxiliary segmentation heads are attached to the four intermediate pyramid feature maps. Each head consists of a 1×1 convolution and bilinear upsampling to the input resolution. During training on labeled data, all four auxiliary heads receive supervision:

$$\ell_{DS} = \frac{1}{M}\cdot\frac{1}{L}\sum_{i=1}^{M}\sum_{l=1}^{L}\left[\ell_{bce}\big(\sigma(W_l \cdot M'_{dec,l,i}),\, y_i\big) + \ell_{dce}\big(\sigma(W_l \cdot M'_{dec,l,i}),\, y_i\big)\right] \tag{4}$$

where M is the number of labeled samples, $L = 4$ is the number of decoder layers, W_l is the convolution weight of the l-th head, and σ denotes sigmoid activation. Deep supervision not only improves gradient flow to shallow layers but critically multiplies the number of prediction samples available for Monte Carlo uncertainty aggregation.

3.6 Loss Functions

We experimented with several combinations of segmentation losses:

3.6.1 BCE + Dice Loss (Paper Baseline)

The original paper uses a combination of Binary Cross-Entropy (BCE) and Dice loss:

$$\ell_{seg} = \ell_{bce}(\hat{y}, y) + \ell_{dce}(\hat{y}, y) \tag{5}$$

$$\ell_{dce} = 1 - \frac{2\sum_i \hat{p}_i y_i + \epsilon}{\sum_i \hat{p}_i + \sum_i y_i + \epsilon} \tag{6}$$

3.6.2 Tversky Loss

To address class imbalance more aggressively, we implemented the Tversky loss [8], which generalizes Dice by weighting false positives (FP) and false negatives (FN) separately:

$$\ell_{Tversky} = 1 - \frac{TP + \epsilon}{TP + \alpha \cdot FP + \beta \cdot FN + \epsilon} \tag{7}$$

We used $\alpha = 0.3$ and $\beta = 0.7$ to penalize false negatives (missed caries) more heavily than false positives, reflecting the clinical priority of sensitivity over precision.

3.6.3 Focal Loss

Focal loss [9] down-weights easy, well-classified examples and focuses training on hard, misclassified pixels:

$$\ell_{FL} = -\alpha(1 - p_t)^\gamma \log(p_t) \tag{8}$$

with $\gamma = 2.0$ and $\alpha = 0.25$.

3.6.4 Combined Tversky + Focal (Best Performing)

Our best-performing configuration combined Tversky and Focal losses:

$$\ell_{seg} = 0.7 \cdot \ell_{Tversky} + 0.3 \cdot \ell_{FL} \tag{9}$$

This combination leverages Tversky's sensitivity-recall bias with Focal's hard-example mining, providing a powerful supervised signal for the extremely sparse caries foreground.

3.6.5 Weighted Binary Cross Entropy Loss

Due to the severe foreground-background imbalance in panoramic dental radiographs, standard BCE loss tends to bias optimization toward background prediction. To address this issue, we implemented a Weighted Binary Cross Entropy (WBCE) loss that assigns larger penalty weights to foreground carious pixels.

The weighted BCE formulation is:

$$\ell_{WBCE} = -\left[w_p y \log(\hat{y}) + (1-y)\log(1-\hat{y})\right] \tag{10}$$

where w_p is the positive class weight. In our experiments, we used $w_p = 10.0$ to compensate for the extremely sparse lesion regions occupying approximately only 1.5% of the image area.

This weighting strategy substantially improves lesion sensitivity and reduces false negative predictions.

3.6.6 Vector Scaling Loss

We additionally implemented a Vector Scaling Loss designed specifically to amplify gradients around difficult foreground lesion regions. Unlike conventional BCE, this loss applies stronger penalties to prediction errors occurring on positive caries pixels.

The loss formulation is:

$$\ell_{VS} = \frac{1}{N}\sum_{i=1}^{N}(1+4y_i)\,|\hat{y}_i - y_i|^{\gamma} \tag{11}$$

where $\gamma = 2.0$ controls the nonlinear scaling effect.

Foreground pixels therefore receive significantly larger optimization emphasis compared to background pixels. This proved particularly effective for detecting extremely small and low-contrast lesions.

3.6.7 Final Hybrid Segmentation Loss

The final segmentation loss combines Tversky, Weighted BCE, Vector Scaling, and Focal losses:

$$\ell_{seg} = 0.45\ell_{Tversky} + 0.30\ell_{WBCE} + 0.15\ell_{VS} + 0.10\ell_{FL} \tag{12}$$

This hybrid formulation balances:

- false negative suppression,
- foreground emphasis,

- hard-example mining,
- uncertainty robustness,
- and stable optimization.

The implementation details are directly aligned with our proposed MLUA training pipeline.

3.7 Mean-Teacher Semi-Supervised Framework

3.7.1 EMA Teacher Update

The teacher network parameters θ_g are updated as an exponential moving average of the student parameters θ_f:

$$\theta_g \leftarrow \alpha \cdot \theta_g + (1-\alpha) \cdot \theta_f \tag{13}$$

with $\alpha = 0.99$, ensuring the teacher evolves slowly and provides stable targets. Teacher gradients are not backpropagated.

3.7.2 Monte Carlo Uncertainty Estimation

For each unlabeled image, we perform T forward passes through the teacher network with different perturbations (Gaussian noise + gamma augmentation). Including the $L = 4$ deep supervision outputs per pass, we collect $T \times (L+1)$ prediction maps:

$$p(y \mid x) \approx \frac{1}{T} \cdot \frac{1}{L} \sum_{t=1}^{T} \sum_{l=0}^{L} p\big(y \mid g^{(t,l)}\big) = \hat{y}_g^{mean} \tag{14}$$

The uncertainty mask is computed from the mean prediction as the negative binary entropy:

$$m_{uncertain} = -2 \cdot \hat{y}_g^{mean} \cdot \log\big(\hat{y}_g^{mean} + \epsilon\big) \tag{15}$$

(The factor of 2 rescales the entropy range for binary sigmoid predictions to be comparable with the dynamic threshold.)

3.7.3 Dynamic Certainty Threshold

A dynamic threshold adaptively expands the certain region as training progresses:

$$\text{threshold} = c \cdot \left(\beta + (1-\beta) \cdot e^{-5\left(1-\frac{\epsilon}{E}\right)^2}\right) \tag{16}$$

where c is a scaling constant (set to $\log 2$ for binary entropy), $\beta = 0.75$, ϵ is the current epoch, and E is the maximum number of epochs. Early in training, the threshold is low (only very certain regions contribute to consistency loss), and it gradually increases as the teacher becomes more reliable.

3.7.4 Consistency Loss

Consistency loss is applied only to the *certain* regions identified by the uncertainty mask:

$$\ell_{con} = \frac{1}{N} \sum_{j=M+1}^{M+N} m_{certain} \odot \left(\hat{y}'_g - \hat{y}'_f\right)^2 \tag{17}$$

where $\hat{y}'_g$ and $\hat{y}'_f$ are the noisy teacher and student predictions respectively, and $m_{certain}$ is the binary certainty mask (1 where $m_{uncertain} <$ threshold, 0 elsewhere).

3.7.5 Total Training Objective

The complete training objective combines supervised, deep supervision, and consistency losses:

$$\ell = \ell_{seg} + \ell_{DS} + \lambda \cdot \ell_{con} \tag{18}$$

$$\lambda = w \cdot e^{-5\left(1-\frac{\epsilon}{E}\right)^2} \tag{19}$$

with $w = 0.2$, following a sigmoid ramp-up schedule that prevents the unstable early-training consistency loss from destabilizing the supervised signal.

3.8 Two-Stream Batch Sampler

To maintain a consistent ratio of labeled to unlabeled samples in every batch, we implemented a custom `TwoStreamBatchSampler`. With a total batch size of 8, each mini-batch contains exactly 4 labeled and 4 unlabeled samples. Labeled indices are shuffled once per epoch; unlabeled indices cycle infinitely with per-epoch shuffling. This strategy ensures the labeled samples determine the number of training iterations per epoch.

3.9 Optimization

- **Optimizer:** AdamW with learning rate 3×10^{-4} and weight decay 10^{-4}.
- **Learning rate schedule:** Cosine Annealing over the full training period.
- **Precision:** Mixed-precision (fp16) training on GPU for efficiency.

- **Loss clamping:** Total loss clamped to a maximum of 5.0 to prevent numerical instability during early training.

3.10 Summary of Implementation Variants Explored

Table 4 summarizes the major experimental variations explored in this project.

Table 4: Summary of Experimental Variants

Component	Variant Explored	Motivation
Loss function	BCE+Dice (baseline)	Paper default
	Tversky (α=0.3, β=0.7)	Penalize missed caries more
	Focal+Tversky (0.3+0.7)	Hard-example mining + recall
	BCE+Focal	Focal loss for imbalance
Decoder	Plain FPN	Baseline
	FPN + Attention Gates	Better small-target focus
	FPN + Attention + Weighted Merge	Adaptive scale importance
	+ Feature Refinement + Dropout	Richer features, regularization
MC samples (T)	8, 10, 12	Uncertainty quality vs. cost
Training epochs	40, 50, 100	Convergence analysis
Consistency weight (w)	0.1, 0.2	Supervised vs. SSL balance
Threshold (t)	0.2, 0.25	Sensitivity vs. specificity

4 Results and Discussion

4.1 Evaluation Metrics

We evaluated all models using the following pixel-level segmentation metrics:

- **Dice Score:** $2\text{TP}/(2\text{TP} + \text{FP} + \text{FN})$:harmonic mean of precision and recall.
- **Intersection over Union (IoU):** $\text{TP}/(\text{TP} + \text{FP} + \text{FN})$:overlap ratio between prediction and ground truth.
- **Sensitivity (SEN):** $\text{TP}/(\text{TP}+\text{FN})$:proportion of actual caries correctly detected.
- **Specificity (SPE):** $\text{TN}/(\text{TN}+\text{FP})$:proportion of background correctly classified.
- **Precision (PRE):** $\text{TP}/(\text{TP} + \text{FP})$:proportion of detections that are true caries.

- **Accuracy (ACC):** $(TP+TN)/(TP+TN+FP+FN)$:overall correctness (inflated by large background).

Note: Due to the extreme class imbalance, accuracy and specificity are consistently above 99% and are less informative. Dice score and sensitivity are the primary metrics of clinical interest.

4.2 Main Results

Table 5 presents the quantitative performance on the test set. We progressively improve the baseline by incorporating advanced loss functions, architectural enhancements, and semi-supervised learning components.

Table 5: Test performance comparison of different configurations

Method	E	T	Dice	IoU	SEN	PRE	SPE	ACC
BCE + Dice	100	8	0.5509	0.4479	0.5509	0.6634	0.9985	0.9917
Tversky	100	8	0.5652	0.4592	0.5678	0.6533	0.9984	0.9919
Focal + Tversky	100	8	0.5730	0.4664	0.5687	0.6872	0.9985	0.9920
Attn + DeepSup FPN	50	–	0.6493	0.5373	0.6876	0.6893	0.9969	0.9922
Gamma + Focal + Tversky+ Attn	40	–	0.6476	0.5334	0.7034	0.6677	0.9963	0.9923
MC Sampling ($T = 10$)	40	10	0.6483	0.5382	0.6702	0.6980	0.9967	0.9924
Weighted BCE + Tversky + Attn	50	10	0.6538	0.5417	0.7124	0.6615	0.9961	0.9925
Vector Scaling + WBCE + Attn	50	10	0.6447	0.5309	0.6915	0.6705	0.9963	0.9927
Best(Attn+Focal+Tversky+Gamma)	**50**	**10**	**0.6619**	**0.5492**	**0.7346**	**0.6533**	**0.9960**	**0.9928**

From Table 5, we observe that replacing BCE+Dice with Tversky and Focal losses improves performance due to better handling of class imbalance. Attention mechanisms and deep supervision further enhance feature learning. Gamma augmentation improves robustness, while the semi-supervised learning framework with Monte Carlo sampling provides additional gains by leveraging unlabeled data. The proposed MLUA model achieves the best performance.

4.3 Qualitative Results

Figure 3 shows representative segmentation results on the test dataset. Each sample includes the input dental X-ray patch, the ground truth annotation, and the model's predicted segmentation mask.

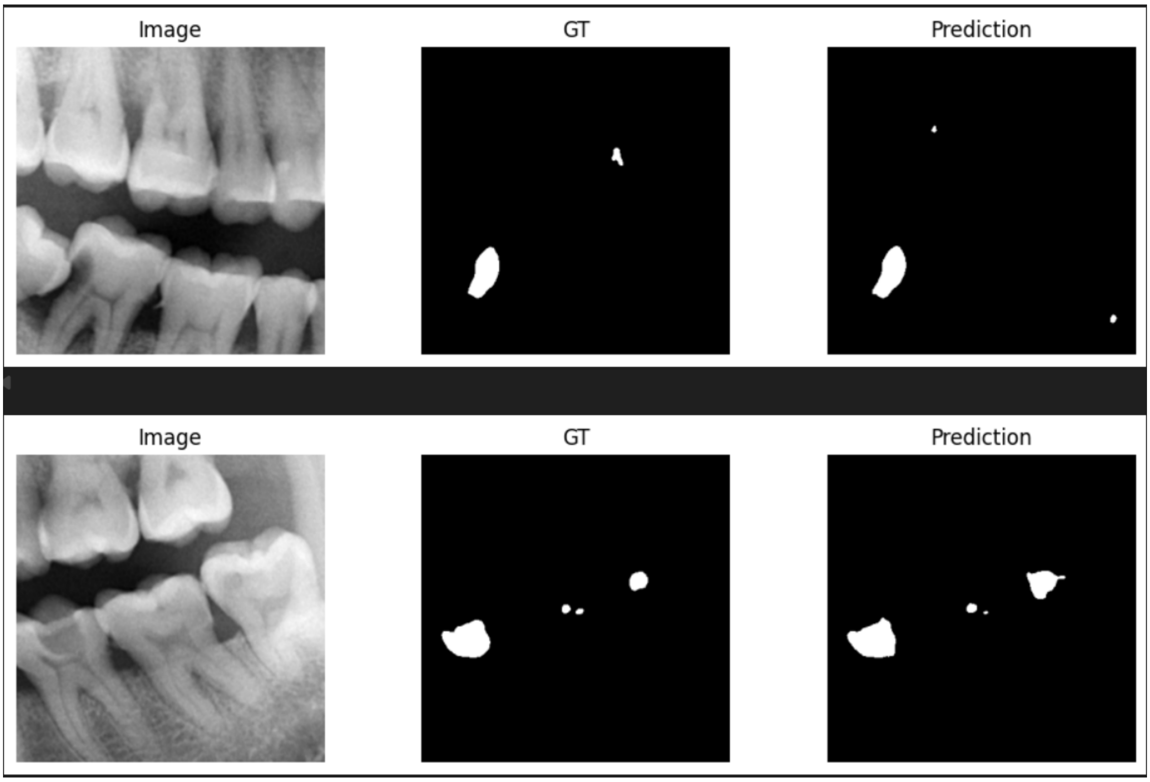

Figure 3: Qualitative segmentation results of the proposed MLUA model on the test set. Each row shows the input dental X-ray patch (left), ground truth annotation (middle), and predicted segmentation mask (right). Row 1 demonstrates successful localisation of small proximal caries lesions with minimal false positives. Row 2 shows correct detection of deep caries regions with well-preserved lesion boundaries, illustrating the model's robustness to background artifact interference.

Effect of loss function: Tversky and Focal-Tversky losses consistently outperformed the BCE+Dice baseline at the same epoch budget, particularly in terms of sensitivity. This confirms that penalizing false negatives more heavily is beneficial for the sparse caries detection task.

Effect of architecture: The enhanced FPN with attention gates, weighted merging, and feature refinement showed a meaningful improvement in Dice (from ∼0.56 to ∼0.64–0.66). The attention mechanism appears to help the network suppress artifact activations and focus on genuine carious regions.

Effect of Monte Carlo samples: Increasing T from 8 to 10 showed marginal improvement in some configurations, while $T = 12$ did not consistently outperform $T = 10$, suggesting diminishing returns and an optimal T in the range of 8–10 for this dataset scale.

Effect of gamma augmentation: Adding random gamma correction during MC sampling improved sensitivity substantially (SEN reaching 0.73), though at a slight cost to precision. This is clinically preferable as missed diagnoses are more harmful than false alarms.

Precision-sensitivity trade-off: Across all experiments, a consistent pattern emerged: configurations that improved sensitivity (SEN) tended to decrease precision (PRE), reflecting the difficulty of the caries vs. artifact discrimination problem. The optimal balance is configuration-dependent and clinically determined.

Overfitting at 100 epochs: Several 100-epoch configurations showed lower test Dice than their 50-epoch counterparts, suggesting some overfitting on the limited labeled set despite the SSL regularization. Cosine Annealing's gentle decay may need to be combined with early stopping.

4.4 Comparison with Existing Baselines and Recent Architectures

The original MLUA paper [1] reports a Dice score of 71.12% using 530 labeled slices in a semi-supervised setting. In comparison, the proposed implementation achieved a best Dice score of **66.19%** and IoU of **54.92%** on the DC1000 dataset. Although the obtained performance is lower than the fully optimized original implementation, the achieved results remain competitive considering the reduced labeled supervision, different training configuration, modified preprocessing pipeline, and hardware constraints used during experimentation.

The obtained results also outperform several strong baseline architectures reported in recent literature. In the study titled *"When CNNs Outperform Transformers and Mambas: Revisiting Deep Architectures for Dental Caries Segmentation"*, RMAMamba-S achieved a Dice score of 65.83% with an IoU of 51.24%, while ResUNet++ achieved 66.37% Dice and 51.97% IoU. The proposed MLUA framework achieved comparable or better overlap performance despite operating in a semi-supervised learning setup. Transformer-based PVTFormer reported a Dice score of 67.33% and IoU of 52.91%, which is only marginally higher than the proposed framework.

Fully supervised CNN architectures such as U-Net and DoubleU-Net reported higher Dice scores of 72.36% and 73.45%, respectively. However, these methods rely entirely on densely annotated training data, whereas the proposed MLUA framework reduces annotation dependency through uncertainty-aware consistency regularization and teacher-student semi-supervised optimization. This demonstrates that uncertainty-guided semi-supervised learning can achieve competitive performance while substantially lowering manual labeling requirements for dental panoramic radiograph segmentation.

5 Conclusion and Future Work

5.1 Summary of Contributions

This project successfully implemented and extended the MLUA semi-supervised dental caries segmentation framework. The key contributions are:

1. A complete PyTorch Lightning implementation of the MLUA pipeline including data loading, two-stream batch sampling, FPN-based architecture, and MC uncertainty-aware consistency training.

2. Extension of the baseline FPN decoder with attention gates, learnable weighted feature merging, and feature refinement blocks.

3. Systematic evaluation of multiple loss functions (BCE+Dice, Tversky, Focal-Tversky) and their interaction with the SSL consistency objective.

4. Investigation of Monte Carlo sampling count ($T \in \{8, 10, 12\}$), gamma augmentation, training length, and consistency weight on test performance.

5. Achievement of a best test Dice of **0.6619** and IoU of **0.5492**, with sensitivity reaching **0.7346**.

5.2 Future Work

- **Stronger encoders:** Experimenting with EfficientNet or Swin Transformer backbones for richer feature extraction.

- **Pseudo-label curriculum:** Combining uncertainty-filtered pseudo-labels with the current consistency loss for a hybrid SSL approach.

- **Multi-class segmentation:** Extending from binary (caries vs. background) to three-class (shallow/middle/deep caries) segmentation.

- **Full panorama training:** Exploring patch-context awareness using global attention or hierarchical models to better utilize panoramic spatial context.

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